# A Framework of Geo-AI for Object Detection and Classification using OGC Standards
# - from Data Collection to Visualization

January 2026

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UN Open GIS
INITIATIVE

# Background

Urban Digital Twin Interoperability Pilot Report

Open Geospatial Consortium

Aug. 2024 – Feb. 2025

Funded By Korea Land and Housing

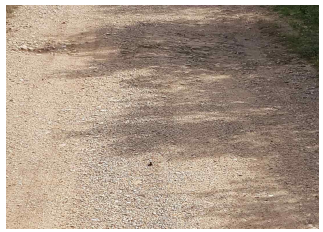
Soheil Sabri Editor

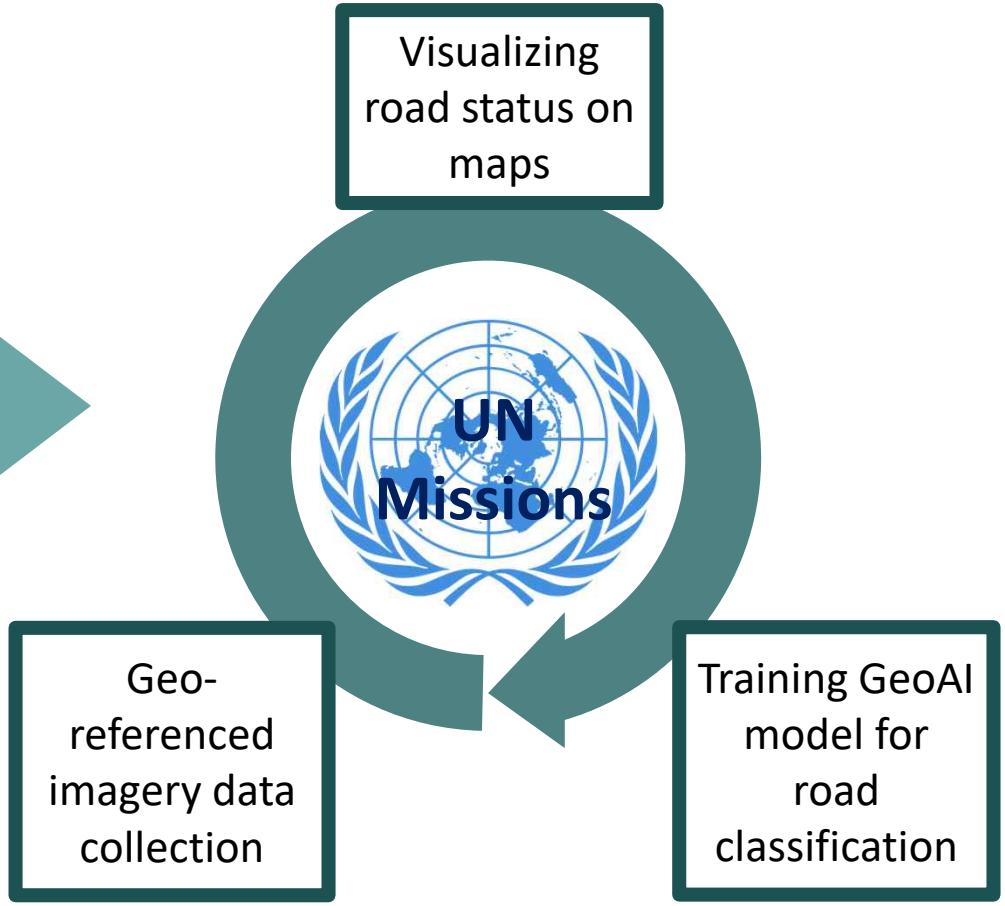
Sina Taghavikish Editor

# Background

UN field missions are largely affected by road conditions such as flooding

Automated Classification on Road Surfaces



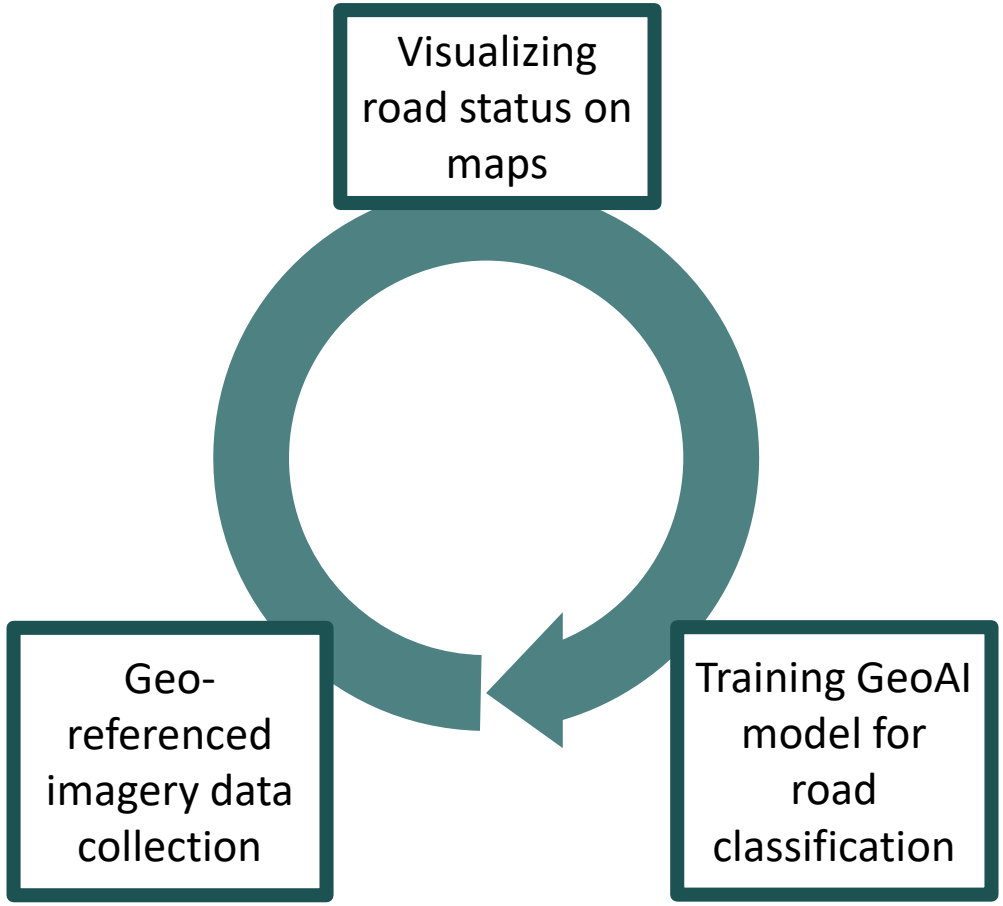


# Goal

Establishment of a **GeoAI Framework** for Road Surface Classification

and
Object Detections

Based on Geospatial Standards:
OGC Standards



# Goal

- Provide a Geo-AI framework for object detection road classification
  - Detect obstacles or classify road surfaces using Geo-AI and
  - Visualize it on web maps (such as OSM)

a road surface

asphalt

paved road

unpaved road

# Sponsors and Contributors - OGC UDTIP project

## Contributors

Sponsors (Users)

Participants (Developers)

LH 한국토지주택공사

UNITED NATIONS

Stakeholder Engagement

Visualization

UDTIP Middleware

Data Providers

Geo-AI

Coordination

Open Geospatial Consortium

Potential Users

Pusan National Univ. and USC

USC University of Southern California

Univ. Central Florida

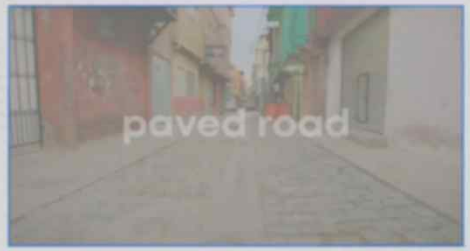
Wuhan Univ.

WiTech

Hochschule für Technik Stuttgart

Helyx

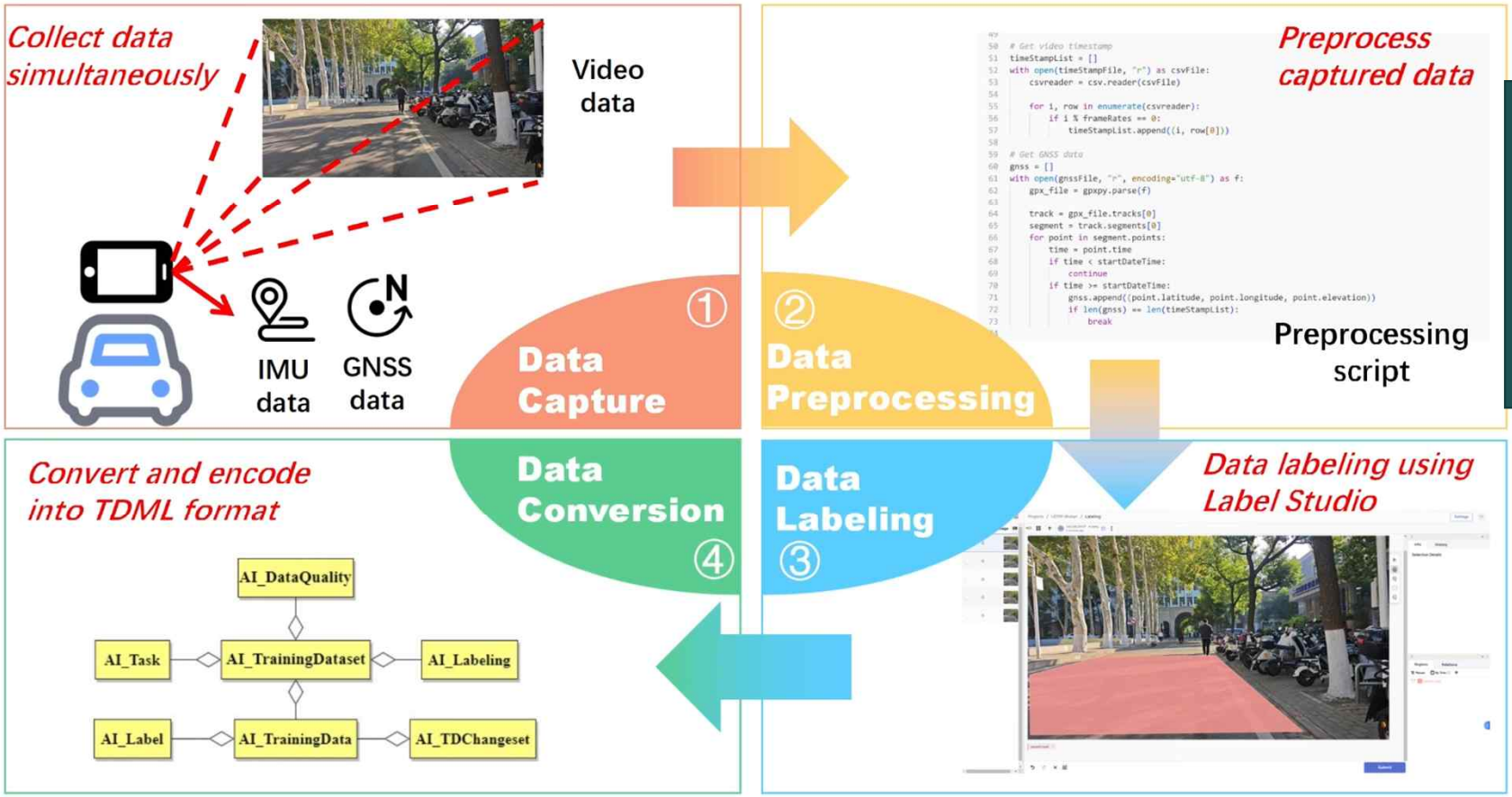
# Overview




Excerpt from UDTIP project video

# A. Geo-tagged Image Collection – Technical Aspects

## Workflow for D101 Camera Imagery Interoperability




Two OGC Standards are employed in data collection phase

- GeoPose
- TrainDML for AI

# A. Geo-tagged Image Collection - Labelling

*RECORDING*

**Location:** UCF Main Campus, Orlando, FL, USA

Date: 9/13/2024 (mm/dd/yyyy)

Start Time: 13:40:34

End Time: 14:10:45

**Total Time:** 30 minutes, and 11 seconds

**Total Distance:** 9.8 km

# A. Geo-tagged Image Collection - OGC GeoPose

**OGC GeoPose Standard** defines geographically-anchored poses

- for each video frame, a json file is defined to specify the location, timestamp, and the orientation such as

```
{
  "id": "camera-001",
  "type": "GeoPose",
  "timestamp": "2025-11-17T09:50:00Z",
  "frame": "EPSG:4979",
  "pose": {
    "position": {
      "lat": 35.3606,
      "lon": 138.7274,
      "h": 3776.0
    },
    "angles": {
      "yaw": 135.0,
      "pitch": -15.0,
      "roll": 3.0
    }
  }
}
```

[source: OGC Testbed-20: Maritime and Aerial Geotagged Video Analysis]

to be included **in OGC TrainDML for AI**

- to provide meta data for training data of AI

# A. Geo-tagged Image Collection - OGC TrainDML for AI

**OGC TrainDML for AI Standard** provides the UML model and encodings for geospatial ML training data

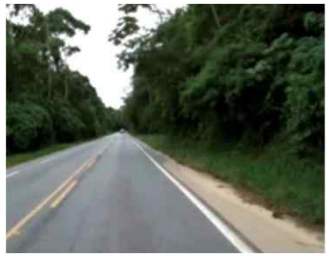
- mainly metadata of AI training data
- extracted from GeoPose data


```
{
    "type": "AI_EOTrainingDataset",
    "id": "D101_example",
    "name": "D101_example",
    "description": "Example dataset for UDTIP D101.",
    "version": "1.0",
    "amountOfTrainingData": 47,
    "createdTime": "2025-01-01",
    "providers": [
        "UDTIP"
    ]
    "classes": [
        {
            "key": "Paved Road",
            "value": null
        },
        {
            "key": "Unpaved Road",
            "value": null
        },
        {
            "key": "Asphalt Road",
            "value": null
        }
    ],
    "numberOfClasses": 3,
    "tasks": [
        {
            "type": "AI_EOTask",
            "id": "whu_example",
        }
    ],
    "data": [
        {
            "type": "AI_EOTrainingData",
            "id": "road_0000",
            "dataURL": [
                "dataset/frame_0000.jpg"
            ],
            "dataTime": "2024-12-01",
            "geopose": {
                "position": {
                    "lon": 114.35413057,
                    "lat": 30.53005352,
                    "h": 19.744
                },
                "quaternion": {
                    "x": -0.6948143326140601,
                    "y": -0.17674656664355012,
                    "z": 0.6955324418081127,
                    "w": -0.04720505021965216
                }
            },
            "labels": [
                {
                    "type": "AI_SceneLabel",
                    "class": "Asphalt Road"
                }
            ]
        },
        ...
    ]
```


*extracted from GeoPose*

# B. Classification and Detection by GeoAI - Goals

- Goal: **To classify road surface conditions** as requested by
  - UN GSC(Global Service Center) and UN VMC (Verification Mission in Colombia)
- as defined by OSM (https://wiki.openstreetmap.org/wiki/Key:surface)
- but initially into Asphalt, Paved, and Unpaved



# B. Classification and Detection by GeoAI - Data Sets

Two data sets

- RTK (Road Traversing Knowledge) data set: public open dataset

(62,662 frame)

(13,697 frame)

(5,702 frame)

<asphalt>

<paved>

<unpaved>

- Total 43min video(34min for train, 9min for test), approx. 77,547 frames total
- Consists **asphalt**, **paved**, **unpaved** dataset
- Asphalt dataset is the largest among the three types of roads

Source: https://towardsdatascience.com/road-surface-classification-150f9874faef

- UN VMC
  - 1012 images

<asphalt>

<paved>

<unpaved>

# B. Classification and Detection by GeoAI - DL Models

## Model Selection

- **ResNet50 [1]**
  - Deep residual network with skip connections to prevent gradient vanishing.
  - Residual connections allow deeper networks to be trained effectively, improving performance over shallow vanilla CNNs.
- **InceptionV3 [2]**
  - Uses factorized convolutions to capture multi-scale features.
  - InceptionV3's ability to extract multi-scale features and its efficient computation make it superior to vanilla CNNs, which lack this capability.
- **ResNet152-v2 [3]** : Deeper version of ResNet with improved architecture
- **DenseNet121 [4]** : Reduces feature redundancy through dense connections
- **EfficientNet_b0 [5]** : Baseline model of efficient neural network architecture scaling


[1] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition" Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770-778

[2] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the Inception Architecture for Computer Vision," Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 2818-2826

[3] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Identity mappings in deep residual networks. European Conference on Computer Vision (ECCV), 630-645.

[4] Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 4700-4708.

[5] Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. International Conference on Machine Learning (ICML), 6105-6114.

# B. Classification and Detection by GeoAI - DL Models

**Accuracy**

| InceptionV3 | Precision | Recall | F1-score |
|---|---|---|---|
| asphalt | 0.8780 | 0.9625 | 0.9183 |
| paved | 0.9305 | 0.8180 | 0.8707 |
| unpaved | 0.9287 | 0.9517 | 0.9401 |
| **Resnet50** | **Precision** | **Recall** | **F1-score** |
| asphalt | 0.9539 | 0.9992 | 0.9760 |
| paved | 1 | 0.9757 | 0.9877 |
| unpaved | 0.9992 | 0.9761 | 0.9875 |
| **Resnet152-v2** | **Precision** | **Recall** | **F1-score** |
| asphalt | 0.9672 | 1 | **0.9833** |
| paved | 1 | 0.9844 | **0.9922** |
| unpaved | 0.9976 | 0.9792 | **0.9883** |
| **Densenet121** | **Precision** | **Recall** | **F1-score** |
| asphalt | 0.9396 | 0.9992 | 0.9685 |
| paved | 1 | 0.9621 | 0.9807 |
| unpaved | 0.9992 | 0.9737 | 0.9863 |
| **Efficientnet_b0** | **Precision** | **Recall** | **F1-score** |
| asphalt | 0.9063 | 1 | 0.9509 |
| paved | 1 | 0.9753 | 0.9875 |
| unpaved | 1 | 0.9214 | 0.9591 |

# C. Visualization of Analyzed Results

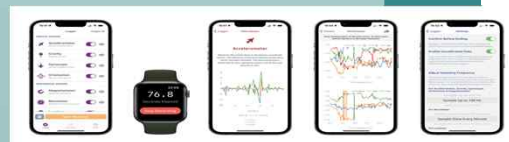
# Open Technologies

**Developed using**

- **open source software**
- **OGC standards**

CVAT

Label Studio



**awesome sensor logger**

OpenCamera Sensors

opengeospatial/**ogcapi-3d-geovolumes**

FFmpeg

PyTorch

CESIUMJS

# Open Source S/W

| Name | Description | License | Site URL |
|---|---|---|---|
| CVAT | Image labeling tool for training data | MIT | https://www.cvat.ai/ |
| Label Studio | Image labeling tool for training data | Apache 2.0 | https://labelstud.io/ |
| Sensor Logger | cross-platform data logger that logs readings from c ommon motion-related sensors on smartphones. | MIT | https://github.com/tszheichoi/awesome-sensor-logger |
| OpenCamera Sensors | Android application for synchronized recording of vi deo and IMU data. It records sensor data (accelero meter, gyroscope, magnetometer) and video with fr ame timestamps synced to the same clock. | GPL 3.0 | https://github.com/MobileRoboticsSkoltech/Op enCamera-Sensors |
| FFMPEG | FFmpeg is a collection of libraries and tools to process multimedia content such as audio, video, subtitles and related metadata. | Mostly LGPL 2.1<br>Some modules are MIT, X11, BSD | https://github.com/FFmpeg/FFmpeg |
| CesiumJS | 3D visualization engine | Apache 2.0 | https://cesium.com/platform/cesiumjs/ |
| Node.js | Node.js is an open-source, cross-platform JavaScript runtime environment. | Mostly MIT, Apache 2.0, and ICU | https://github.com/nodejs/node |
| Express.js | Fast, unopinionated, minimalist web framework for Node.js. | MIT | https://github.com/expressjs/express |

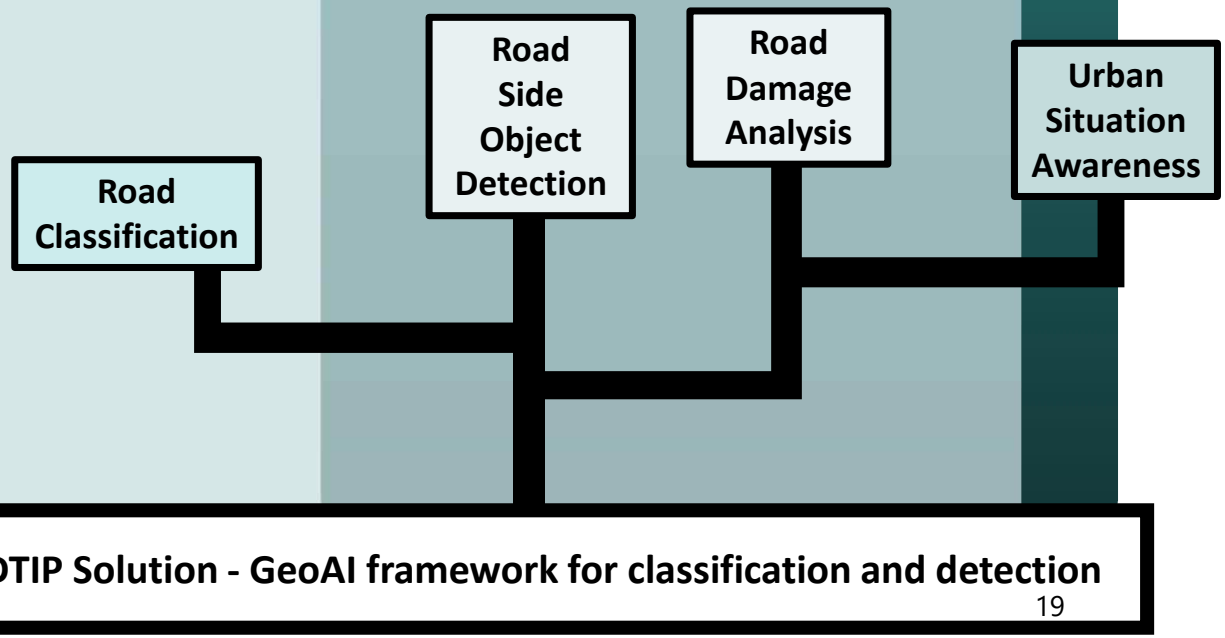
# Extensibility

**Highly extensible**

- initial road surface types: asphalt, paved, and unpaved
  → may include other types such as unpaved and mud
  → only to modify TrainDML and apply proper DL model.

```
"classes": [
    {
        "key": "Paved Road",
        "value": null
    },
    {
        "key": "Unpaved Road",
        "value": null
    },
    {
        "key": "Asphalt Road",
        "value": null
    }
],
"numberOfClasses": 3,
```

**extend it**

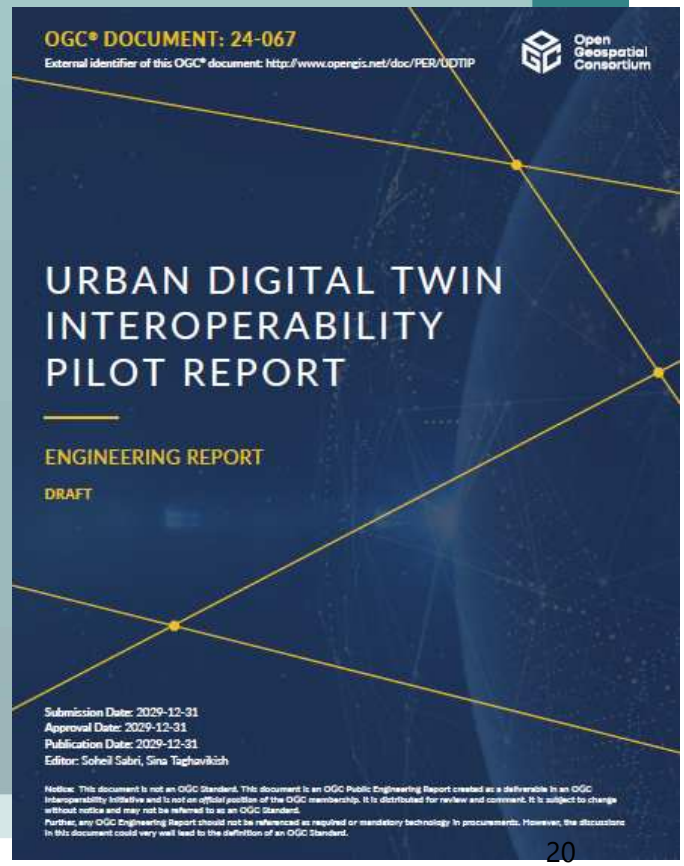


# Summary

## GeoAI framework for road classification and detection

- Developed as a part of OGC Pilot Project UDTIP funded by Korea LH
  - initially to support UN field missions

- Using OGC Standards for Interoperability
- Detail information is available at OGC Engineering Report https://docs.ogc.org/per/24-067r1.html
- If you have any questions or suggestions, please contact
  Ki-Joune Li (lik@pnu.edu)
  Sina Taghavikish (staghavikish@ogc.org)



**Thank You**

**Any Questions?**